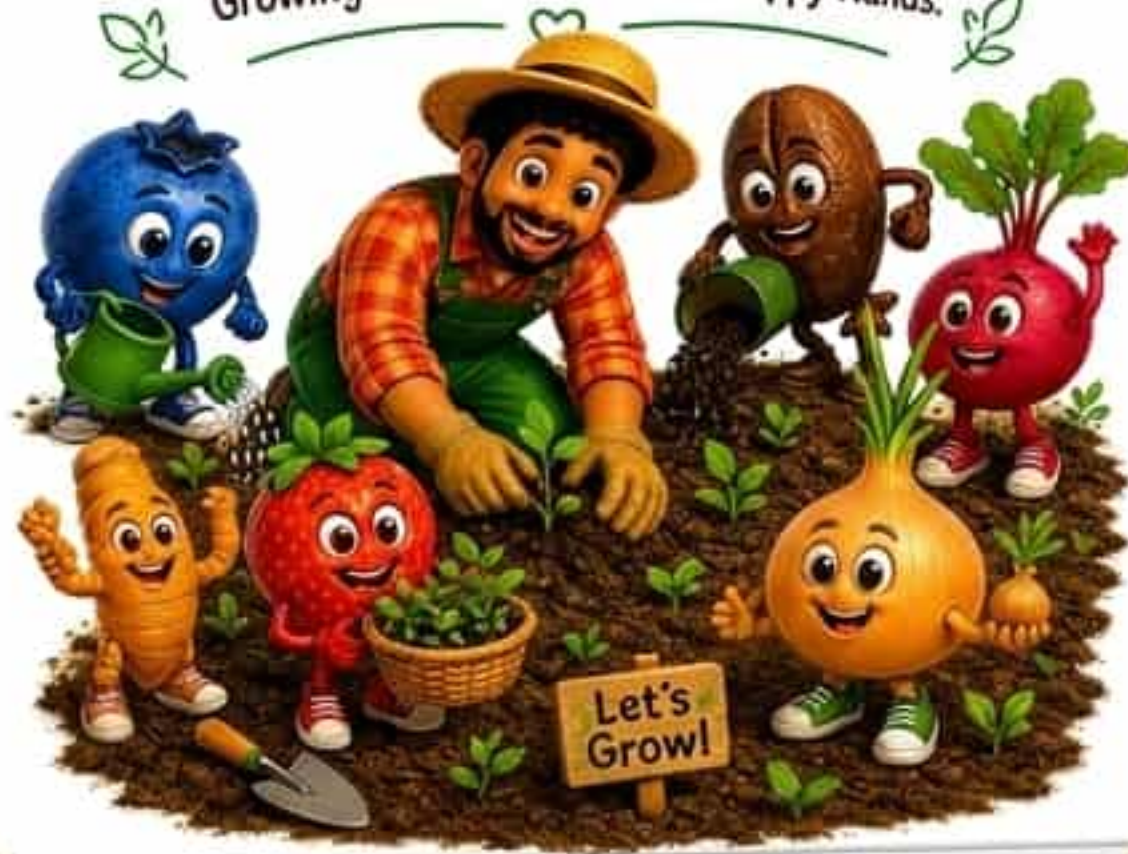


The **Roots** & **Shoots** Explorer's Field Journal

Coach your favorite soils needs to the organization design.



LESSON TWO **ROOTS & SHOOTS** Growing Food. Healthy Soil. Happy Hands.



Learn now w fons with
Good Food. Strong
Bodies. Happy Hearts.

LESSON ONE **ROOTS & SHOOTS** Good Food. Strong Bodie. Happy Hearts.

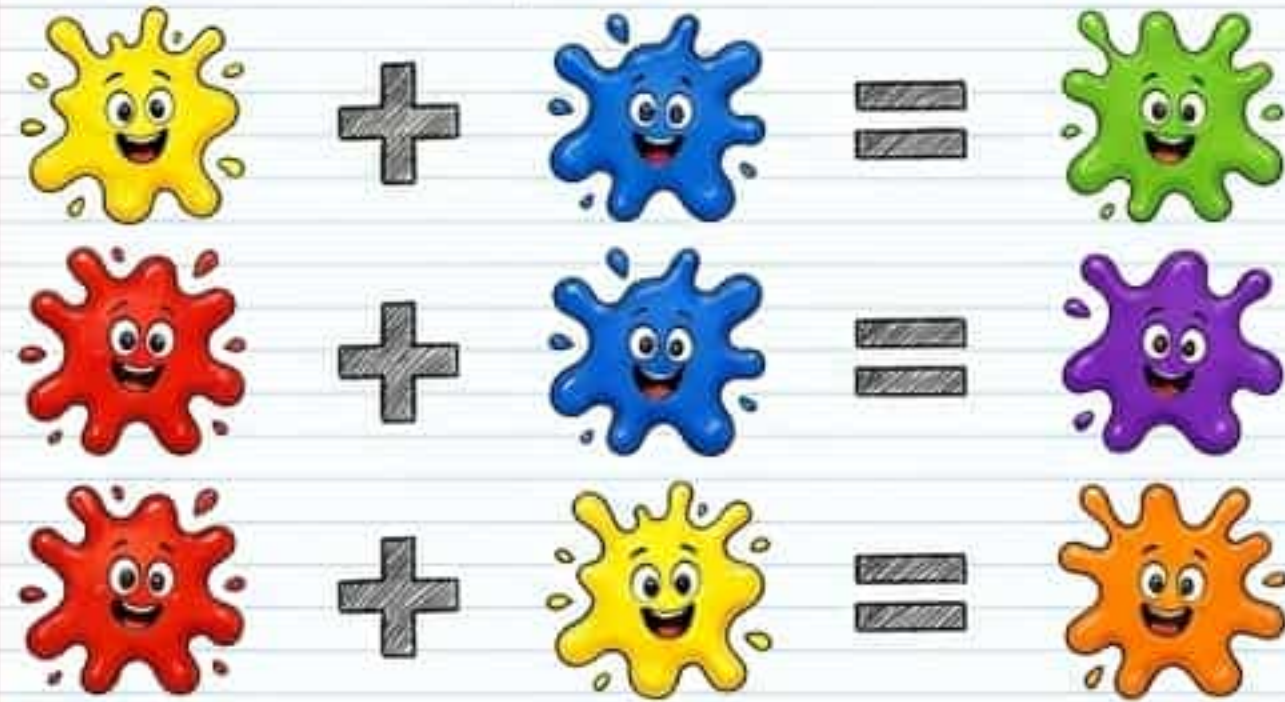
Everything Starts in the Garden Grocery Store



The garden is a grocery store providing good food for strong bodies. Every plant relies on the secret world hidden just beneath its roots.

Nature's Alchemy in the Colour Kitchen

Alchemy Equation Grid



When foundational ingredients mix together, they create entirely new forms of life and colour in the garden!

The Compost Decision Tree



Kitchen Leftovers

Did this come from a plant or nature?



YES



NO



COMPOST:
Food for the soil.

Super Soil requires the right organic matter!

TRASH: Cannot be broken down by our garden heroes.



The Elevator Drop Through the Earth's Crust



Organic Matter

Leaves, twigs, and plant bits break down to feed the soil.

Topsoil

Dark, rich soil full of nutrients. Great for plant roots!

Subsoil

Lighter soil with fewer nutrients. Some small rocks and minerals.

Small Rocks & Pebbles

Broken-down pieces of rock.

Large Rocks

Bigger rocks broken down over time.

Bedrock

Solid rock layer. The foundational base layer.



★ Geology Fact: It takes thousands of years for solid rock to turn into topsoil!

The Ultimate Soil Showdown

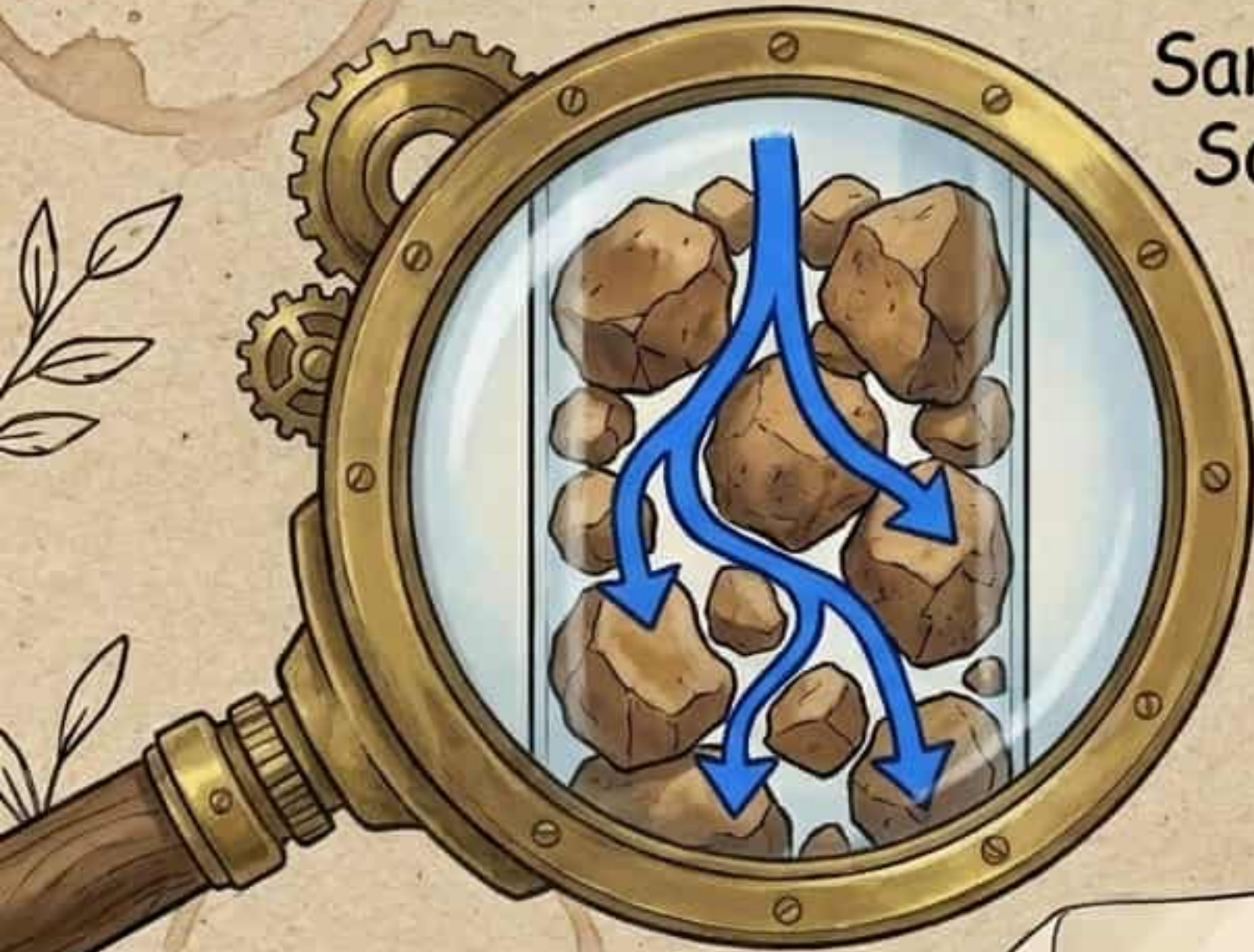
Competitor	Texture	Water Retention	Nutrient Level	Best For
Sandy Soil 	Gritty			
Clay Soil 	Sticky			
Silt Soil 	Soft like flour			
Loam Soil 	Crumbly & Rich			
Peaty Soil 	Spongy			
Chalky Soil 	Stony			

The absolute best for garden life!

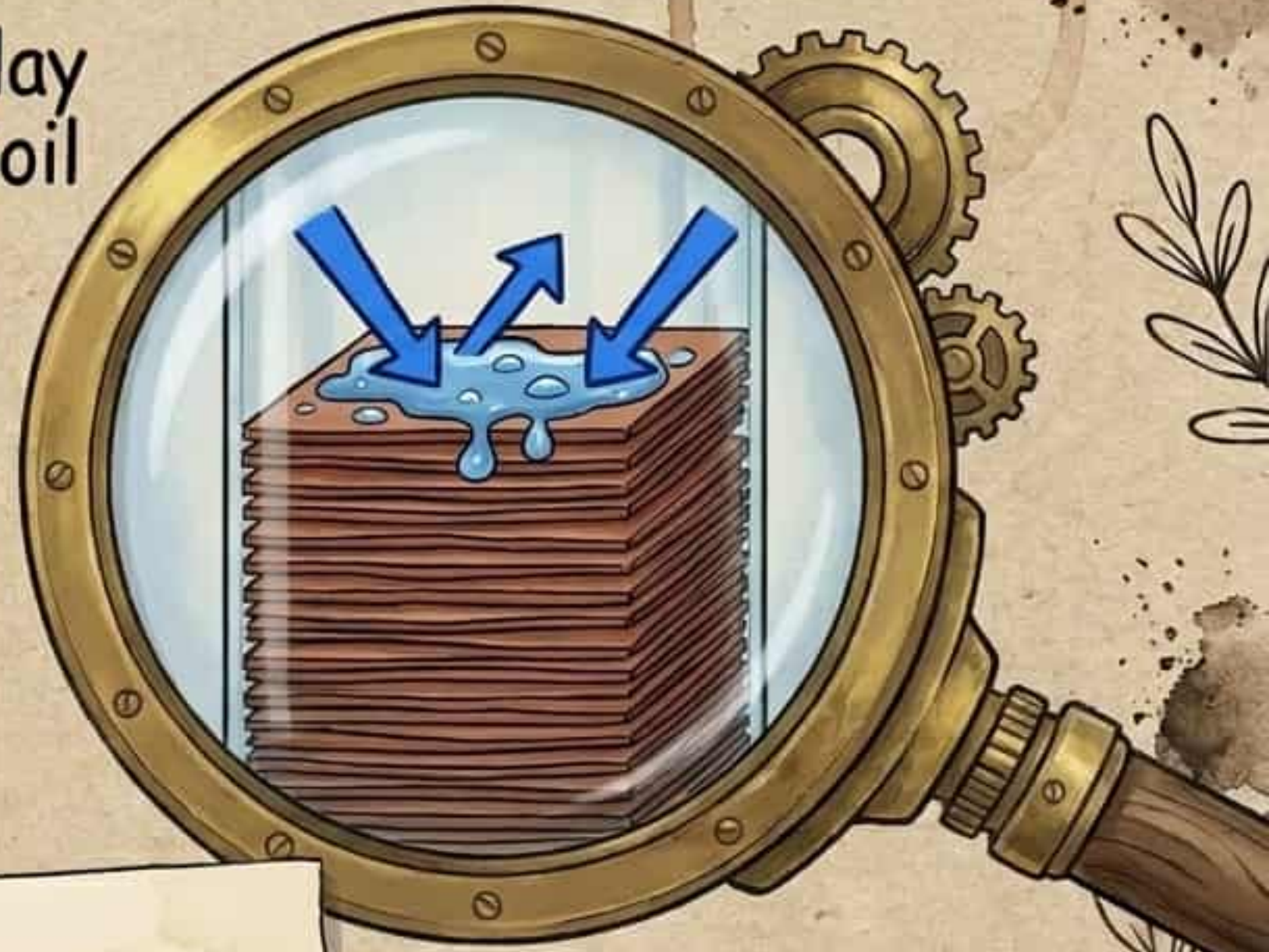
Micro-Architecture Explains How Soil Behaves

Sandy
Soil

Clay
Soil



Large particles create big gaps. Water falls right through.



Tightly packed flat plates leave no room for air. Water gets trapped.

Loam works best because it mixes the boulders (sand) with the plates (clay), creating the perfect balance of airflow and water storage!



The Macroscopic Soil Safari



Earthworms:
The Tunnelers – They dig tunnels, aerate the soil, and help nutrients mix in.



Beetles:
The Decomposers – Help decompose larger plant matter.



Pill Bugs (Roly-Polies):
The Recyclers – Break down dead leaves and return nutrients to the soil.



Springtails:
The Leaf Munchers – Tiny bugs that help break down dead leaves.



Centipedes:
The Balancers – They break down dead material and keep the soil balanced.

Soil is **ALIVE!** These tiny visible heroes work together to prepare the ground for the invisible kingdom.

The Invisible Kingdom: Meet the Microbes

Bacteria (The Blue Rods): The fastest decomposers! They break down simple sugars and proteins.

Protozoa (The Amoebas): Microscopic predators that eat bacteria and release nutrients as they go.

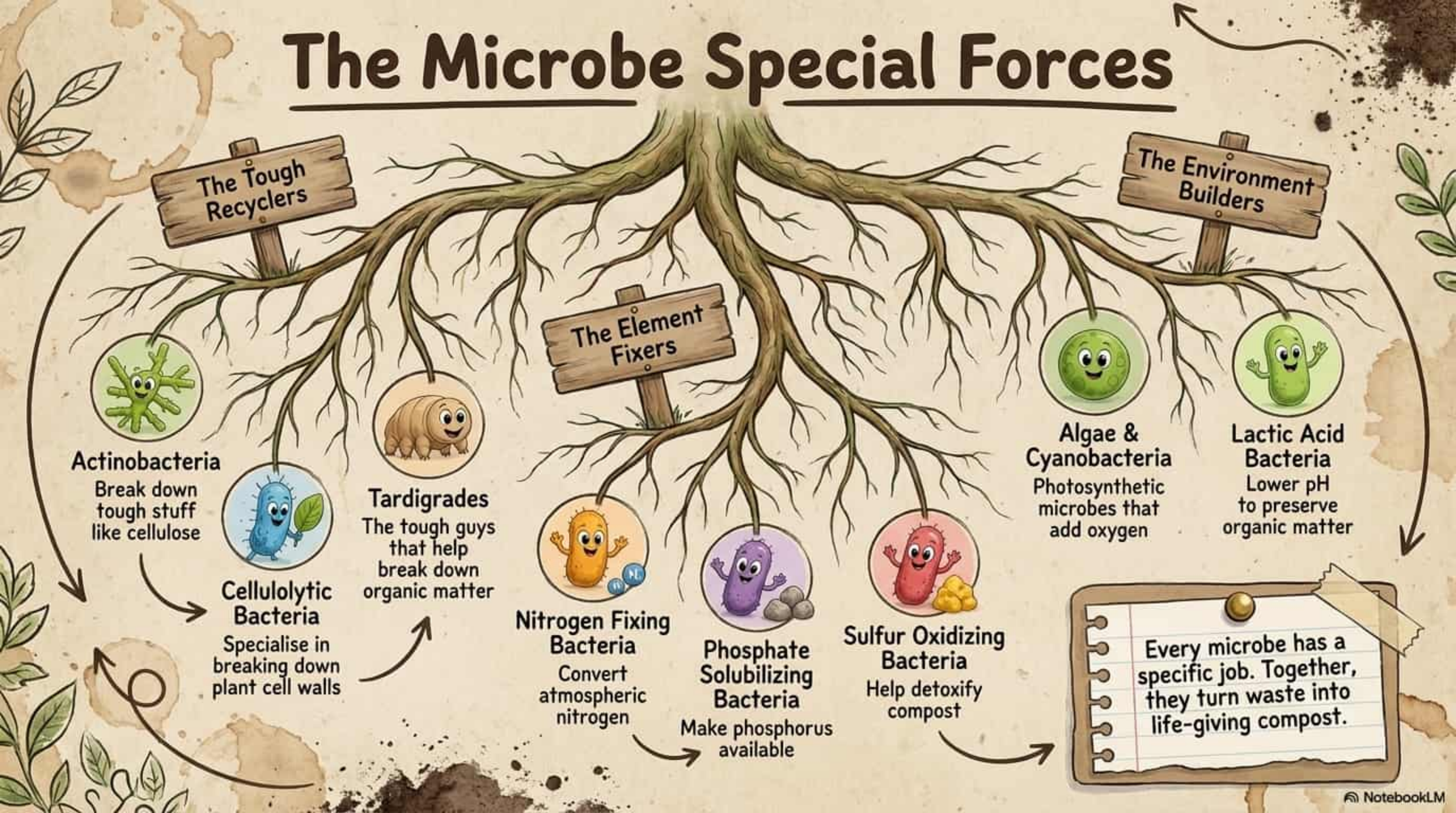
Fungi (The Mushrooms): Nature's ultimate recyclers. They break down tough plant fibres like lignin.

Nematodes (The Worms): Microscopic roundworms that help control bacterial populations.

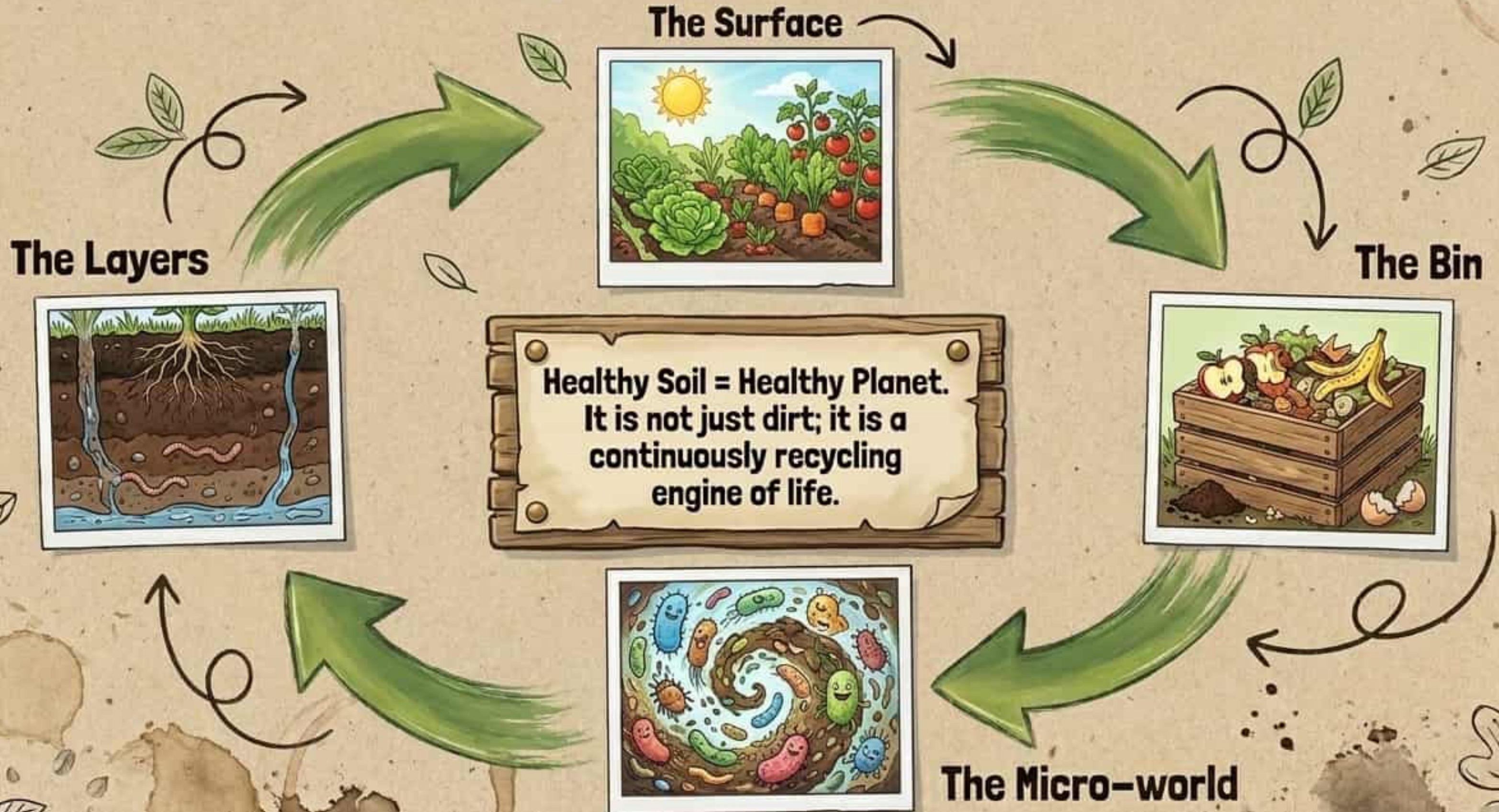
They may be invisible, but they are the true rulers of the garden!



The Microbe Special Forces



The Looping Engine of Garden Life



Field Notes & The Wishing Well



My favourite soil superhero is...

One thing I will look for in my garden is...

To help the microscopic kingdom, I will...

.....
.....
.....



Throw a coin in the wishing well and remember: great things grow from the ground up. Keep exploring!